

The ball dentable property for Bochner L^p -spaces

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Abstract

Basu and Seal prove that $L^p(\mu, X)$ having the ball dentable property implies that X has the ball dentable property, and ask whether the converse holds. We prove the converse for $1 < p < \infty$ and finite nonzero μ . Thus, in this range, $L^p(\mu, X)$ has BDP if and only if X has BDP. Combined with the scalar $L^1[0, 1]$ endpoint counterexample already recorded in the run, this gives the BDP exponent split for the Bochner converse: false at $p = 1$, true for $1 < p < \infty$.

1 Source Question

The source paper introduces the ball dentable property (BDP): a Banach space X has BDP if, for every $\varepsilon > 0$, the unit ball B_X has a slice of diameter less than ε .

In Section 3, the source proposition labelled `bochner bdp` proves:

$$L^p(\mu, X) \text{ has BDP} \implies X \text{ has BDP}$$

for $1 \leq p < \infty$ and finite μ . Immediately after the proof, the authors write that they do not know whether the converse is true in general. The relevant source passage is:

$$S(B_{L^p(\mu, X)}, f^*, \alpha) \cap S(B_{L^p(\mu, X)}, g^*, \alpha) = \emptyset.$$

$$a(S(B_{L^p(\mu, X)}, f^*, \alpha)) \geq \|g - h\|_p = \left(\sum_{i=1}^n \|y_i - z_i\|^p \mu(A_i) \right)^{1/p}$$

is a contradiction.

rk 3.3. We do not know whether the converse of the
 1. However, converse of Proposition 3.1, fails for $p =$
 μ is Lebesgue measure on $[0, 1]$ (which is simply denot

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quote the following result from [GGMS]

osition 4.1 [GGMS] X is strongly regular if and only if

The $p = 1$ converse is already false, even for $X = \mathbb{R}$ and Lebesgue measure, because every slice of the unit ball of non-atomic L^1 has diameter two. The present note proves the positive $1 < p < \infty$ half for BDP.

2 Idea of the Proof

Let $S(B_X, x^*, \eta)$ be a small slice of B_X . We lift it to a slice of $B_{L^p(\mu, X)}$ by the constant dual function

$$F(f) = \int_{\Omega} x^*(f(t)) d\mu(t)$$

after normalizing μ to be a probability measure.

The key point is specific to $1 < p < \infty$. In scalar L^p , the functional $h \mapsto \int h$ strongly exposes the constant function 1 on the positive part of the unit ball. Thus if $F(f)$ is close to 1, then $\|f(\cdot)\|$ is close to 1 in L^p , and the nonnegative defect

$$\|f(t)\| - x^*(f(t))$$

has small integral. Except on a set of small L^p -mass, the normalized vector $f(t)/\|f(t)\|$ must lie in the original small slice of B_X . Two functions in the lifted slice are therefore pointwise close on most of Ω , and the exceptional parts have small L^p -norm.

3 Proof

We use the notation

$$S(B_X, x^*, \eta) = \{x \in B_X : x^*(x) > 1 - \eta\},$$

where $x^* \in S_{X^*}$ and $\eta > 0$.

Lemma 1 (Lifting a small BDP slice). *Let (Ω, Σ, μ) be a probability space, let $1 < p < \infty$, and let $x^* \in S_{X^*}$. Suppose that*

$$S = S(B_X, x^*, \eta)$$

has diameter d . For every $\gamma > 0$ there is $\alpha > 0$ such that the slice

$$\mathcal{S}_\alpha = \left\{ f \in B_{L^p(\mu, X)} : \int_\Omega x^*(f(t)) d\mu(t) > 1 - \alpha \right\}$$

has diameter at most $d + \gamma$.

Proof. It is enough to prove the following sequential assertion: if $\alpha_n \downarrow 0$ and $f_n, g_n \in B_{L^p(\mu, X)}$ satisfy

$$\int x^*(f_n) d\mu > 1 - \alpha_n, \quad \int x^*(g_n) d\mu > 1 - \alpha_n,$$

then

$$\limsup_{n \rightarrow \infty} \|f_n - g_n\|_{L^p(\mu, X)} \leq d.$$

Indeed, failure of the lemma for some $\gamma > 0$ would give such a sequence with all distances $> d + \gamma$.

Put $r_n(t) = \|f_n(t)\|$ and $s_n(t) = \|g_n(t)\|$. Since

$$\int x^*(f_n) d\mu \leq \int r_n d\mu \leq \|r_n\|_{L^p} \leq 1,$$

we have $\int r_n d\mu \rightarrow 1$ and $\|r_n\|_{L^p} \rightarrow 1$. The scalar space $L^p(\mu)$ is uniformly convex, so the constant function 1 is strongly exposed in $B_{L^p(\mu)}$ by the functional $h \mapsto \int h d\mu$. Hence

$$r_n \rightarrow 1 \quad \text{in } L^p(\mu).$$

The same argument gives $s_n \rightarrow 1$ in $L^p(\mu)$. Moreover,

$$\int_\Omega (r_n(t) - x^*(f_n(t))) d\mu(t) \rightarrow 0, \quad \int_\Omega (s_n(t) - x^*(g_n(t))) d\mu(t) \rightarrow 0.$$

Define the bad sets

$$A_n = \{t : r_n(t) = 0\} \cup \left\{ t : r_n(t) > 0 \text{ and } x^*\left(\frac{f_n(t)}{r_n(t)}\right) \leq 1 - \eta \right\},$$

and similarly

$$B_n = \{t : s_n(t) = 0\} \cup \left\{ t : s_n(t) > 0 \text{ and } x^*\left(\frac{g_n(t)}{s_n(t)}\right) \leq 1 - \eta \right\}.$$

On the part of A_n where $r_n > 0$ we have

$$r_n(t) - x^*(f_n(t)) = r_n(t) \left(1 - x^*\left(\frac{f_n(t)}{r_n(t)}\right) \right) \geq \eta r_n(t),$$

so $\int_{A_n} r_n d\mu \rightarrow 0$, while the zero part of A_n has measure at most $\|r_n - 1\|_p^p$. Since $r_n \rightarrow 1$ in L^p , this implies $\mu(A_n) \rightarrow 0$ and then

$$\int_{A_n} r_n^p d\mu \rightarrow 0.$$

For example, $A_n \cap \{r_n > 1/2\}$ has measure at most $2 \int_{A_n} r_n d\mu$, while $A_n \cap \{r_n \leq 1/2\}$ has measure controlled by $2^p \|r_n - 1\|_p^p$; after this, $r_n^p \leq 2^{p-1}(|r_n - 1|^p + 1)$ gives the displayed convergence. The same reasoning gives

$$\mu(B_n) \rightarrow 0, \quad \int_{B_n} s_n^p d\mu \rightarrow 0.$$

Let $C_n = \Omega \setminus (A_n \cup B_n)$. On C_n , both r_n and s_n are positive, and the normalized vectors

$$u_n(t) = \frac{f_n(t)}{r_n(t)}, \quad v_n(t) = \frac{g_n(t)}{s_n(t)}$$

belong to the slice S . Hence $\|u_n(t) - v_n(t)\| \leq d$ on C_n . Therefore

$$\begin{aligned} \|(f_n - g_n)\mathbf{1}_{C_n}\|_p &\leq \|(r_n - s_n)\mathbf{1}_{C_n}\|_p + d \|s_n \mathbf{1}_{C_n}\|_p \\ &\leq \|r_n - 1\|_p + \|s_n - 1\|_p + d. \end{aligned}$$

The right-hand side has limsup at most d .

It remains to check that the complement contributes nothing in the limit. Since $r_n \rightarrow 1$ and $s_n \rightarrow 1$ in L^p , and since $\mu(A_n \cup B_n) \rightarrow 0$, we have

$$\int_{A_n \cup B_n} r_n^p d\mu \rightarrow 0, \quad \int_{A_n \cup B_n} s_n^p d\mu \rightarrow 0.$$

Thus

$$\|(f_n - g_n)\mathbf{1}_{\Omega \setminus C_n}\|_p \leq \|r_n \mathbf{1}_{A_n \cup B_n}\|_p + \|s_n \mathbf{1}_{A_n \cup B_n}\|_p \rightarrow 0.$$

Combining the two estimates proves

$$\limsup_n \|f_n - g_n\|_p \leq d,$$

as required. $\square$

Theorem 1. *Let (Ω, Σ, μ) be a finite nonzero measure space, let $1 < p < \infty$, and let X be a Banach space. If X has BDP, then $L^p(\mu, X)$ has BDP.*

Proof. Since replacing μ by the probability measure $\mu/\mu(\Omega)$ only rescales the L^p -norm by a constant, we may assume that μ is a probability measure.

Fix $\varepsilon > 0$. Since X has BDP, choose $x^* \in S_{X^*}$ and $\eta > 0$ such that the slice

$$S(B_X, x^*, \eta)$$

has diameter less than $\varepsilon/2$. Apply Lemma 1 with $\gamma = \varepsilon/2$. The resulting slice $\mathcal{S}_\alpha$ of $B_{L^p(\mu, X)}$ has diameter less than ε . Therefore $L^p(\mu, X)$ has BDP. $\square$

Corollary 1. *Let (Ω, Σ, μ) be a finite nonzero measure space and $1 < p < \infty$. Then*

$$L^p(\mu, X) \text{ has BDP} \iff X \text{ has BDP.}$$

Proof. The forward implication is the proposition labelled **bochner bdp** in Basu–Seal. The reverse implication is Theorem 1. $\square$

4 Endpoint and Scope

The theorem is sharp in the exponent. At $p = 1$, the converse fails over the Lebesgue interval with $X = \mathbb{R}$: the space X has BDP, while every slice of the unit ball of scalar $L^1[0, 1]$ has diameter two. This endpoint counterexample is recorded separately in the run packet

2011.14591_11_scalar_bdp_converse_counterexample.

This packet does not address the analogous BHP converse. The proof above uses a slice and the strong exposure of the scalar L^p unit ball; it does not convert an arbitrary small relatively weakly open subset of B_X into a small relatively weakly open subset of $B_{L^p(\mu, X)}$.

5 Novelty Check

The cheap run indexes were searched for arXiv:2108.02908, arXiv:2011.14591, BDP, BHP, Lebesgue-Bochner, Ball Dentable Property, and the converse language from the source. The existing $p = 1$ endpoint packet was found and is cited above; it explicitly left the $1 < p < \infty$ positive stability direction outside its scope.

Web searches for exact phrases including “Ball Dentable Property L^p Bochner”, “BDP Lebesgue-Bochner $p > 1$ ”, and the source title returned the source paper but did not locate a separate explicit answer to the $1 < p < \infty$ BDP converse.

6 References

- S. Basu and S. Seal, *Small Combination of Slices, Dentability and Stability Results Of Small Diameter Properties In Banach Spaces*, arXiv:2108.02908.
- S. Basu and S. Seal, *Stability results of small diameter properties in Banach spaces*, arXiv:2011.14591. This is the related arXiv record used by the existing endpoint packet.